

Asia Semiconductors & Global Memory

MediaTek Inc

Rating

Outperform

Price Target

2454.TT

5,500.00 TWD



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MediaTek: ASIC is changing quickly - 4 updates to stay abreast

We summarize four major updates to help investors stay abreast of the ASIC market.

Growing at a pace faster than GPUs. GPU demand is still growing, but in line with expectations. The recent demand surprise has come mostly from ASICs, as inference & agentic workloads favor the efficiency delivered by ASICs and TSMC, etc., allocate more incremental capacity accordingly. We forecast the XPU ASIC market (including HBM but excluding CPU, merchant networking) to enjoy a 4-5x increase to US\$250-300B in 2028.

High memory prices favor ASIC but make sizing ASIC market difficult. We believe high memory prices prompt hyperscalers to use more ASICs to mitigate the burden. Moreover, as Broadcom includes HBM in its revenue & we define our ASIC market accordingly, the changes to memory & thus HBM pricing will distort the size of the market.

Anthropic & OpenAI are emerging as two major “customers”. Both are extending beyond AI models to hardware & silicon. OpenAI's Jalapeño marks the first example. Anthropic is now using TPUs, but likely with its own racks & should develop its own ASICs later. We believe MediaTek is gaining share at Google, but winning Anthropic & OpenAI allows Broadcom to grow as well.

More suppliers & segmentations, but a much bigger market. We forecast Google will stay as the largest customer, esp. when TPUs sold externally are included too. Amazon appears “unfairly” small, but mostly because it captures much of the profit internally. Google is soliciting proposals from more suppliers for future production needs, likely as a result of scale and hence different projects & service levels for different segments. Competition will intensify technically, but a much bigger market will make many suppliers grow.

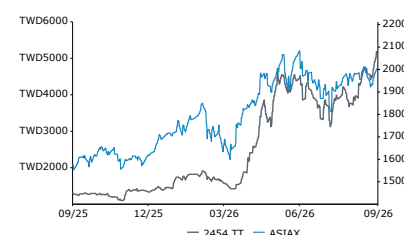
Investment Implications

We rate MediaTek **Outperform**. Our 1-year price target is **NT\$5,500.00**

Close Date	24 Sep 2026			
2454.TT Close Price (TWD)	5,285.00			
Price Target (TWD)	5,500.00			
Upside/(Downside)	4%			
52-Week Range	5,480.00/1,130.00			
ASIAX	1,995.50			
FYE	Dec			
Div Yield	1.0%			
Market Cap (TWD) (B)	8,482.17			
EV (TWD) (B)	8,333.67			
Performance	YTD	1M	6M	12M
Absolute (%)	269.6	40.4	226.2	297.4
ASIAX (%)	22.0	2.8	20.1	26.0
Relative (%)	247.6	37.6	206.2	271.4

Source: Bloomberg, Bernstein estimates and analysis.

Price Performance, 1YR



Reported EPS	F25A	F26E	F27E	Financials	F25A	F26E	F27E	CAGR	Valuation Metrics	F25A	F26E	F27E
2454.TT (TWD)	66.17	68.52	169.35	Revenues (M)	595,966	676,804	1,184*	40.9%	Reported P/E (x)	79.9	77.1	31.2
				Gross Margin (%)	47.5	45.9	44.5	--	Div Yield (%)	1.0	1.0	1.4
				Operating Margin (%)	17.4	16.6	24.7	--	Adjusted P/E (x)	79.9	77.1	31.2
				Net Income Margin (%)	17.8	16.3	23.0	--	EV/EBITDA (x)	65.9	59.9	26.0
				ROE (%)	26.4	25.8	51.7	--	EV/EBIT (x)	80.5	74.0	28.5

*Values shown in billions; Source: Bloomberg, Bernstein estimates and analysis.

See the Disclosure Appendix of this report for required disclosures, analyst certifications and other important information. Alternatively, visit our [Global Research Disclosure Website](#).

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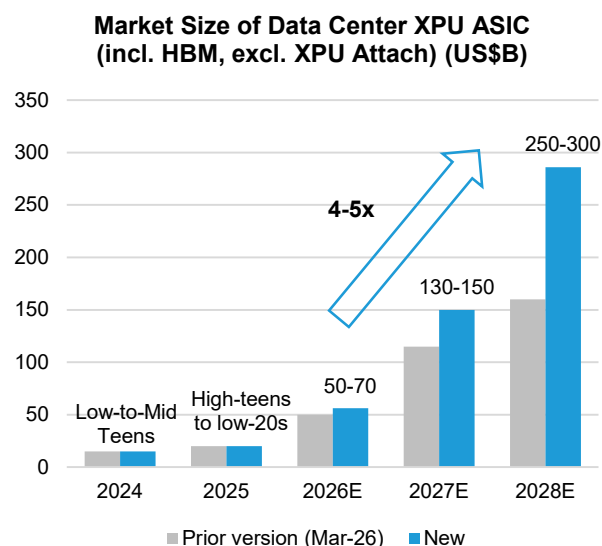
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DETAILS

1) ASIC is growing quickly, at a pace faster than GPU.

- We recently took up our forecast for the XPU (i.e. accelerator) ASIC market significantly ([Exhibit 1](#)). The market enjoyed a 4- to 5- fold increase from 2024 to 2026. Going forward, AI investment is increasingly tilting from development to monetization. Inference & agentic workload thus surges, & efficiency (vs. sheer performance) of compute is becoming critical to scale the deployment size up. We believe ASIC shines in efficiency, and as a result, we forecast the XPU ASIC market to enjoy another 4- to 5-fold increase in the next 2 years, to US\$250-300B in 2028.
- We highlight this does not mean GPU market is becoming smaller. We expect the GPU market to grow as well, but in line with expectations. The recent demand surprise has been however much more from ASIC, and we believe TSMC & other upstream suppliers are also allocating much more incremental capacity to ASIC accordingly.
- Please also note CPU ASIC market is growing quickly additionally. From a low base, this market will remain much smaller than the XPU ASIC market. Our forecast above hence does not include it yet, but the market certainly merits investor attention. We will also try to include the CPU ASIC market in our forecast later.

EXHIBIT 1: We forecast the XPU ASIC market size to enjoy another 4- to 5-fold increase in the next 2 years, to US \$250-300B in 2028.



*Our definition for data center XPU ASIC does not include merchant silicon (e.g. networking chips), nor custom CPU. For the ASIC vendors that exclude HBM from their revenues, we estimate the HBM value and add it to this forecast.

Source: Bernstein estimates (for all years) and analysis

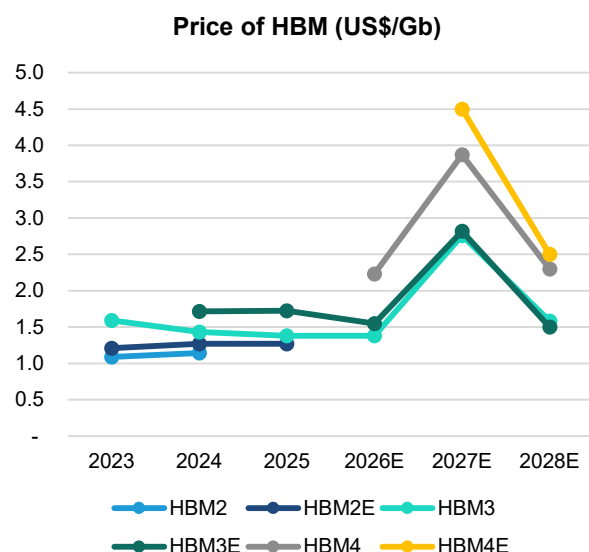
2) High memory prices benefit ASIC but also make HBM an uncertainty when sizing the ASIC market.

- In addition to the increasing tilt to efficiency, high memory prices, we believe, also favor ASIC. AI data centers need to save costs elsewhere to mitigate the high memory prices, and that may explain why we picked up signals of ASIC driving most of the incremental silicon demand in the past few quarters.
- Please also note the forecast above includes the value of HBM, even when some ASIC vendors don't include that in their revenues. MediaTek's TPU projects for Google (GOOGL.US, covered by Bernstein US Internet team) are an example. Though HBM is packaged inside these XPUs, it is Google sourcing HBM directly from memory suppliers. The value of HBM therefore is not part of the revenue recognized by MediaTek. For these projects, we estimate the value of HBM & include it in the

forecast above so that investors can see the full picture. This also makes a like-for-like comparison with GPU easier, as GPU suppliers' revenues & hence GPU market size also have the value of HBM.

- This further implies the size of our forecast is sensitive to the changes of HBM pricing. Though the contracts & price agreements of HBM tend to last longer (usually yearly) & suppliers indeed consider the strategic value of HBM beyond simply profitability, HBM shares wafer capacity with conventional DRAM, and hence needs to adjust its price too in order to win capacity during capacity shortage. How HBM price evolves therefore will affect our projection. [Exhibit 2](#) summarizes the HBM price assumptions embedded in this forecast above & readers will find we forecast the price to "normalize" in 2028. The "normalization" assumption for 2028 makes our forecast more conservative than consensus, implying that the XPU ASIC market would be bigger than our forecast if you believe in the HBM price expected by consensus. For companies including HBM value in their revenues, investors should also note HBM price naturally will affect their revenue guidance, unless they lock in the price through long-term contracts.

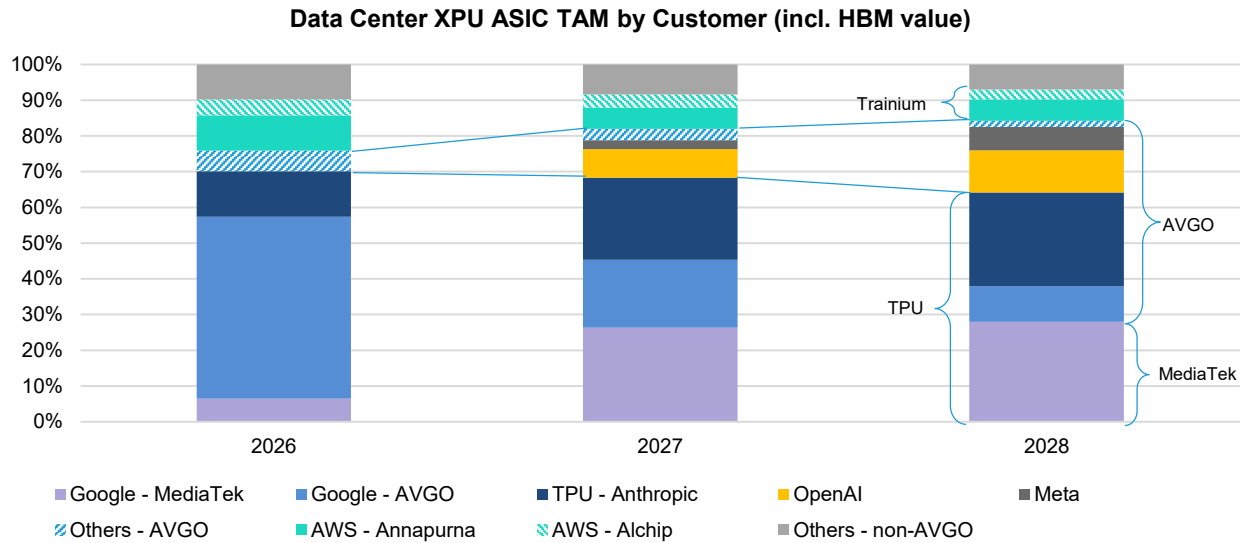
EXHIBIT 2: The HBM assumptions embedded in our ASIC TAM indicate the price should "normalize" in 2028.



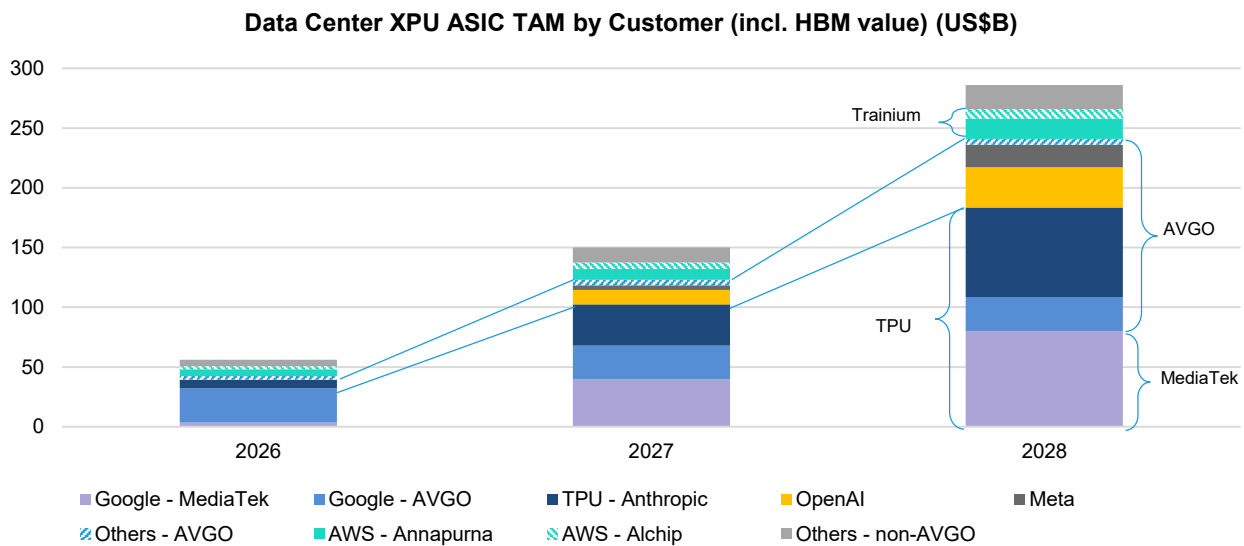
Source: Company reports, TrendForce, Bernstein estimates and analysis

3) "Customer" base is broadening, with Anthropic & OpenAI emerging rapidly but needing more help.

- From merely two AI model companies and simply users of AI compute power, Anthropic and OpenAI (both not listed) are extending their reach upstream as the knowledge & control in silicon is essential to the progress of their frontier models & the scale & efficiency of serving their rapidly expanding user base. OpenAI's Jalapeño marks the first example. Anthropic is becoming a big user of TPUs from Google, but the company is probably working with Broadcom (AVGO.US, covered by Bernstein US Semi team) to run those TPUs into racks different from those used by Google internally. Taking a step further, Anthropic is also building semiconductor expertise internally (for example [a report](#) & [a job post](#)), & we won't be surprised that Anthropic later becomes a direct customer of ASIC service too. However, unless they jump start through M&As, we believe it will take a process for them to accumulate sufficient experience, and the help of ASIC service providers will be necessary for them to walk through it.
- To map out all these developments, we estimate & summarize the end users of the XPU ASIC market in [Exhibit 3](#) & [Exhibit 4](#). These exhibits further show how ASIC service providers are mapped to these end users and how TPU is split between Anthropic & Google's internal use. Clearly Anthropic is becoming a very important end user. So is OpenAI, through its own Jalapeño and other chips.
- Looking at these exhibits but from the supplier perspective, readers will find the rapid growth of MediaTek & how the wins at Anthropic, OpenAI, etc. allow Broadcom to grow at the same time.

EXHIBIT 3: The “customer” base of data center XPU ASIC is broadening, with Anthropic & OpenAI emerging rapidly.

Source: Bernstein estimates (for all years) and analysis

EXHIBIT 4: The expansion of TAM reveals the rapid growth of MediaTek & how the wins at Anthropic, OpenAI, etc., allow Broadcom to grow at the same time.

Source: Bernstein estimates (for all years) and analysis

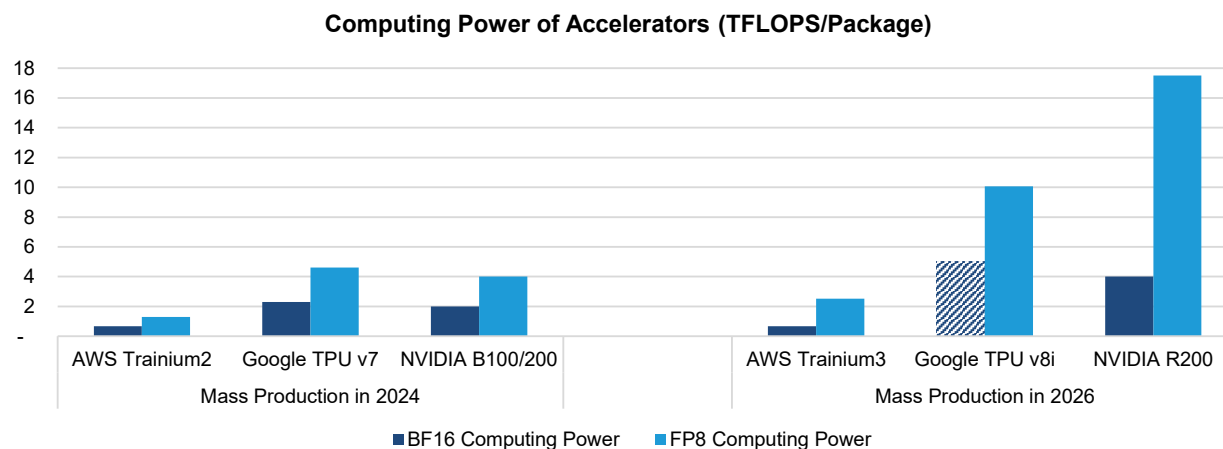
4) Amazon & Google are experienced customers, & continue to work on scale & efficiency. More suppliers will enter the market but segmentation may emerge. Most importantly, the market is big enough for many suppliers to grow.

- Exhibit 3 & Exhibit 4 also show that Google should remain the bigger "customer" of ASIC, especially when the external use of its TPU is also considered. Google is now deploying its 8th generation of TPUs, and the development of its 9th generation has been contracted and ongoing for a while. Google is thus very experienced, & that together with the expanding scale likely call for chips optimized for different workloads & more ASIC service suppliers. That explains why we believe Marvell (MRVL.US, not covered) was awarded by Google a project similar to the LPU (Language Process Unit) in which NVIDIA (NVDA.US, covered by Bernstein US Semi team)/Groq also leverage SRAM to minimize latency for the latency-sensitive workload. This may also explain why we believe that Google is exploring many possible suppliers for the v10 generation, and possibly for

multiple chips. Investors may worry about competition, but we find different competencies of different suppliers & different needs of different workloads should lead to segmentation. More importantly, the demand expansion will be fast enough to make many suppliers grow.

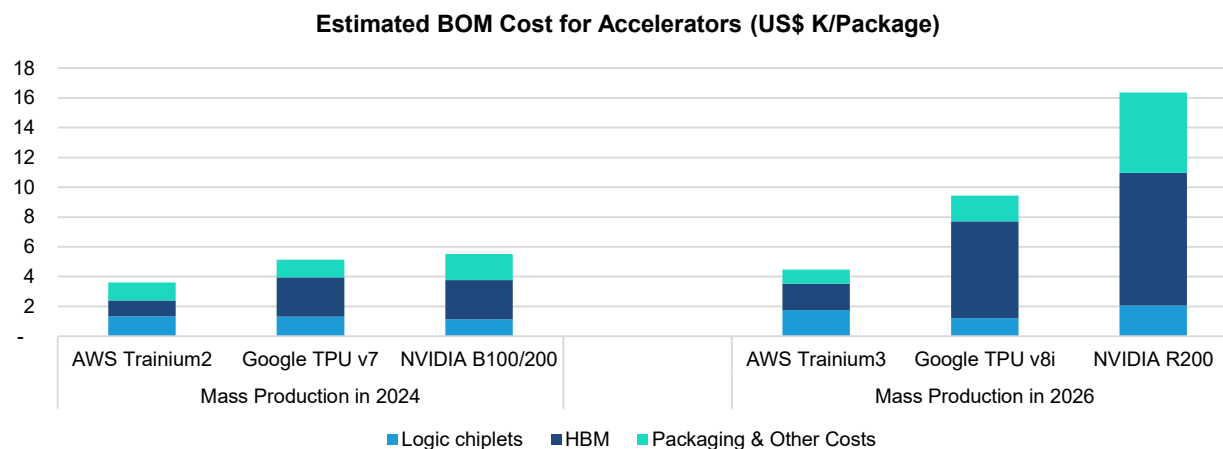
- Amazon (AMZN.US, covered by Bernstein US Internet team) is another experienced ASIC customer, and actually emphasizes efficiency even more than Google. As shown in [Exhibit 5](#) and [Exhibit 6](#), their ASICs are often lighter on hardware specifications. The acquisition of Annapurna in 2015 boosted Amazon's internal capabilities so that they usually retain external help only for easier parts (e.g. backend) of chip designs. Amazon also controls supply chain more, and procures most of the silicon directly from TSMC even in the projects that external ASIC service providers participate. This makes Amazon appear unfairly small in [Exhibit 3](#) & [Exhibit 4](#), as a lot of profits is captured by Amazon itself and is thus not part of our ASIC market definition. Conversely, this also means that Amazon will be much bigger when we look at the quantities of ASIC chips.
- We see the possibility that Google may become as "efficient" as Amazon one day, capturing more value internally & leaving less to external providers. For exactly the same reason, Anthropic, OpenAI, Microsoft (MSFT.US, covered by Bernstein US Software team), Meta (META.US, covered by Bernstein US Internet team), etc. likely will reduce the use of the full services of the US vendors, and shift more engagements to MediaTek and other Asian providers. Investors should be mindful of the lumpiness as the results of big customers changing their suppliers, but at the same time don't have to panic as all these are a natural evolution commonly seen in ASIC industry. As long as the market is big enough, the long-term investment thesis should not change.

EXHIBIT 5: **AWS's XPU ASICs are often lighter on hardware specifications...**



*The computing power in BF16 of Google TPU v8i is our estimate.
Source: AWS, Google, NVIDIA, Bernstein estimates and analysis

EXHIBIT 6: ...and require lower BOM cost to build.



Source: AWS, Google, NVIDIA, Bernstein estimates (for all years) and analysis

We remain confident in MediaTek and rate it Outperform.

- In summary, we reiterate our positive stance on MediaTek. TPU v8 & v9 projects should deliver earnings notably above consensus. Competition in v10 may turn more intense, but MediaTek is well positioned to defend its share, and also to win projects beyond TPU. **Outperform.**

APPENDIX - FINANCIAL FORECASTS

EXHIBIT 7: MediaTek Financial Forecasts

NT\$ in Millions	2026				2027				2025	2026E	2027E	2028E
	Q1	Q2	Q3E	Q4E	Q1E	Q2E	Q3E	Q4E				
Income Statement												
Net Sales	149,151	152,183	158,728	216,743	259,676	305,607	317,038	301,639	595,966	676,804	1,183,960	2,156,293
Gross Profit	69,055	70,308	74,366	97,039	114,833	134,582	141,380	136,545	283,080	310,767	527,339	924,430
SG&A Expense	7,819	8,079	8,571	9,814	8,718	8,526	9,121	9,913	31,304	34,284	36,278	38,448
R&D Expense	38,345	39,361	39,682	46,531	46,243	49,455	49,715	53,081	148,306	163,919	198,494	236,710
EBIT	22,891	22,868	26,113	40,693	59,872	76,600	82,544	73,551	103,470	112,565	292,568	649,273
EBITDA	28,892	29,708	32,943	47,597	66,854	83,649	89,654	80,717	126,444	139,139	320,874	679,660
Non-operating Income	4,129	5,094	3,319	3,433	6,038	6,888	9,322	9,206	21,418	15,974	31,454	57,664
Pretax Income	27,019	27,962	29,432	44,126	65,910	83,488	91,866	82,758	124,888	128,539	324,021	706,937
Tax	2,643	3,357	4,856	7,501	10,546	13,358	14,699	13,241	18,770	18,357	51,843	113,110
Net Income	24,154	24,335	24,354	36,295	54,866	69,503	76,479	68,871	105,319	109,138	269,719	588,418
Earnings Per Share (NT\$)	15.17	15.28	15.29	22.79	34.45	43.64	48.02	43.24	66.17	68.52	169.35	369.45
Margin Analysis												
Gross Profit	46.3%	46.2%	46.9%	44.8%	44.2%	44.0%	44.6%	45.3%	47.5%	45.9%	44.5%	42.9%
EBIT	15.3%	15.0%	16.5%	18.8%	23.1%	25.1%	26.0%	24.4%	17.4%	16.6%	24.7%	30.1%
EBITDA	19.4%	19.5%	20.8%	22.0%	25.7%	27.4%	28.3%	26.8%	21.2%	20.6%	27.1%	31.5%
Pretax Income	18.1%	18.4%	18.5%	20.4%	25.4%	27.3%	29.0%	27.4%	21.0%	19.0%	27.4%	32.8%
Net Income	16.2%	16.0%	15.3%	16.7%	21.1%	22.7%	24.1%	22.8%	17.7%	16.1%	22.8%	27.3%
SG&A as % of Sales	5.2%	5.3%	5.4%	4.5%	3.4%	2.8%	2.9%	3.3%	5.3%	5.1%	3.1%	1.8%
R&D as % of Sales	25.7%	25.9%	25.0%	21.5%	17.8%	16.2%	15.7%	17.6%	24.9%	24.2%	16.8%	11.0%
Sequential Growth												
Net Sales	-0.7%	2.0%	4.3%	36.5%	19.8%	17.7%	3.7%	-4.9%	12.3%	13.6%	74.9%	82.1%
Gross Profit	-0.3%	1.8%	5.8%	30.5%	18.3%	17.2%	5.1%	-3.4%	7.5%	9.8%	69.7%	75.3%
EBIT	4.8%	-0.1%	14.2%	55.8%	47.1%	27.9%	7.8%	-10.9%	1.0%	8.8%	159.9%	121.9%
EBITDA	4.4%	2.8%	10.9%	44.5%	40.5%	25.1%	7.2%	-10.0%	2.5%	10.0%	130.6%	111.8%
Pretax	-0.5%	3.5%	5.3%	49.9%	49.4%	26.7%	10.0%	-9.9%	4.5%	2.9%	152.1%	118.2%
Net Income	5.4%	0.7%	0.1%	49.0%	51.2%	26.7%	10.0%	-9.9%	-1.0%	3.6%	147.1%	118.2%
Earnings Per Share	5.3%	0.7%	0.1%	49.0%	51.2%	26.7%	10.0%	-9.9%	-1.2%	3.6%	147.1%	118.2%
Balance Sheet												
Cash and Cash Equivalents									235,290	373,250	563,548	1,040,266
Accounts Receivable									62,121	117,795	196,721	402,526
Inventory									67,235	156,136	215,340	459,916
Net Fixed Assets									60,427	58,584	48,038	49,995
Total Assets									743,785	1,078,778	1,415,727	2,414,079
Accounts Payable									48,710	78,068	107,243	240,966
Total Debt									13,730	155,385	153,979	139,206
Total Equity									409,195	455,096	609,360	939,504
Cash Flow												
Cash Flow From Operations									162,793	121,494	292,386	706,558
Cash Flow From Investments									(37,754)	(29,524)	(18,189)	(32,782)
Cash Flow From Financing									(87,675)	44,043	(83,900)	(197,057)
Depreciation & Amortization									22,974	26,575	28,306	30,387
Capex									(15,059)	(16,447)	(17,759)	(32,344)
Key Financial Metrics												
ROE									26.4%	25.8%	51.7%	77.4%
ROA									14.3%	10.2%	19.2%	24.6%
Debt / Equity									3.4%	34.1%	25.3%	14.8%
Net Debt / Equity									-54.1%	-47.9%	-67.2%	-95.9%
Debt / Capital									3.2%	25.5%	20.2%	12.9%
Dividend Per Share (NT\$)									53.5	50.7	73.9	165.2

Source: Company reports, Bernstein estimates and analysis

BERNSTEIN TICKER TABLE

Ticker	Rating	Cur	24 Sep 2026	Price Target	TTM	Reported EPS				Reported P/E (x)		
			Closing Price		Rel. Perf.	Cur	2025A	2026E	2027E	2025A	2026E	2027E
2454.TT (MediaTek)	O	TWD	5,285.00	5,500.00	271.4%	TWD	66.17	68.52	169.35	79.9	77.1	31.2
ASIAAX			1,995.50									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

Source: Bloomberg, Bernstein estimates and analysis.

DISCLOSURE APPENDIX

I. REQUIRED DISCLOSURES

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VALUATION METHODOLOGY**MediaTek Inc**

We set a NT\$5,500 1-year price target using a P/E multiple of 18x against our forward Q5-Q8 EPS estimate of NT\$306.

RISKS**MediaTek Inc**

There are a number of downside risks to our MediaTek price target: (1) Less than expected shipment of ASICs & (2) Stronger competitive pressure, including from customers' internal team, and (3) Weaker smartphone demand

RATINGS DEFINITIONS, BENCHMARKS AND DISTRIBUTION**EQUITY RATINGS DEFINITIONS****Bernstein brand**

The Bernstein brand rates stocks based on forecasts of relative performance for the next 12 months versus the S&P 500 for stocks listed on the U.S. and Canadian exchanges, versus the Bloomberg Europe Developed Markets Large and Mid Cap Price Return Index EUR (EDME) for stocks listed on the European exchanges and emerging markets exchanges outside of the Asia Pacific region, versus the Bloomberg Japan Large and Mid Cap Price Return Index USD (JPL) for stocks listed on the Japanese exchanges, and versus the Bloomberg Asia ex-Japan Large and Mid Cap Price Return Index (ASIAX) for stocks listed on the Asian (ex-Japan) exchanges -unless otherwise specified.

The Bernstein brand has three categories of ratings:

- Outperform: Stock will outpace the market index by more than 15 pp
- Market-Perform: Stock will perform in line with the market index to within +/- 15 pp
- Underperform: Stock will trail the performance of the market index by more than 15 pp

Coverage Suspended: Coverage of a company under the Bernstein research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Not Covered (NC) denotes companies that are not under coverage.

Bernstein brand stock ratings are based on a 12-month time horizon.

Autonomous brand – common stocks

The Autonomous brand rates common stocks as indicated below. As our benchmarks we use the Bloomberg Europe 600 Financials Price Return Index (E600BK) and Bloomberg Europe Dev Mkt Financials Large and Mid Cap Price Ret Index EUR (EDMFI) index for developed European banks and Payments, the Bloomberg Europe 600 Insurance Price Return Index (E600IN) for European insurers, the S&P 500 and S&P Financials for US banks and Payments coverage, S5LIFE for US Insurance, the S&P Insurance Select Industry (SPSIINS) for US Non-Life Insurers coverage, the Bloomberg Emerging Markets Financials Large, Mid and Small Cap Price Return Index (EMLSF) for emerging market banks and insurers and Payments, and the Bloomberg Japan Financials Large & Mid Cap Price Return Index (JPFILM) for Japanese Financials. Ratings are stated relative to the sector (not the market).

The Autonomous brand has three categories of common stock ratings:

- Outperform (OP): Stock will outpace the relevant index by more than 10 pp
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- Underperform (UP): Stock will trail the performance of the relevant index by more than 10 pp

Coverage Suspended: Coverage of a company under the Autonomous research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Those denoted as 'Feature' (e.g., Feature Outperform FOP, Feature Under Outperform FUP) are our core ideas.

Not Covered (NC) denotes companies that are not under coverage.

Autonomous brand common stock ratings are based on a 12-month time horizon.

Autonomous brand – preferred stocks

The Autonomous brand has three categories of preferred stock ratings:

- Outperform (OP): The total return of the preferred instrument is expected to outperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- Neutral (N): The total return of the preferred instrument is expected to perform in line with preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- Underperform (UP): The total return of the preferred instrument is expected to underperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.

Autonomous preferred stock ratings are based on a 6-month time horizon.

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The report may also include reference(s) to published opinions by other Autonomous or Bernstein analysts covering the equity securities of the issuer(s) referenced herein. Please note an investment recommendation for credit instruments published by the author(s) of this report may differ from the published view of the analyst covering equity securities for the issuer(s) contained in this report and vice versa.

CREDIT RATINGS DEFINITIONS

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- **Credit Neutral (C-N):** The total return of the Reference Credit Instrument is expected to perform in line with the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.
- **Credit Underperform (C-UP):** The total return of the Reference Credit Instrument is expected to underperform the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.

Autonomous credit ratings are based on a 6-month time horizon.

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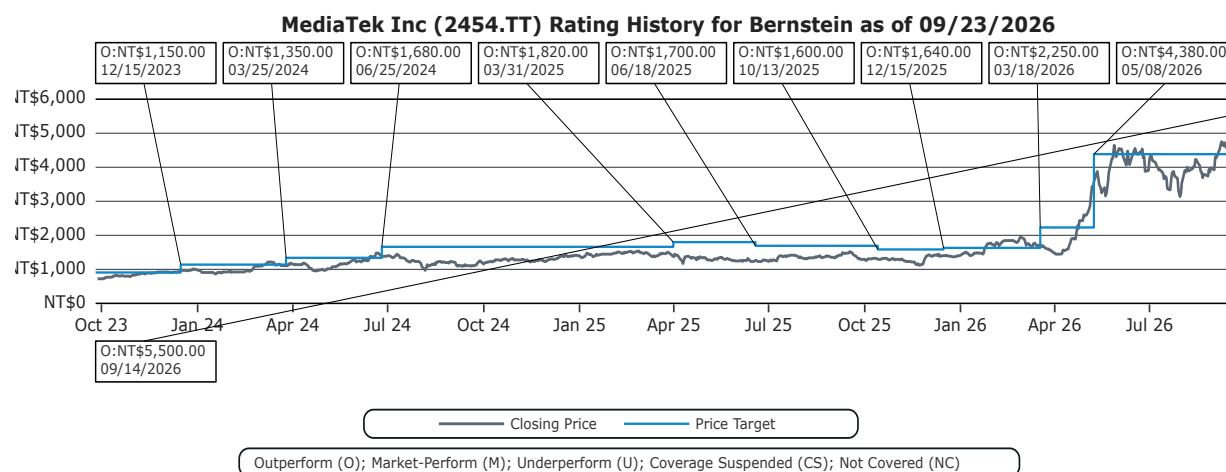
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